

**Astro 331 SPRING 2026 Syllabus (rev 2)**

|         | Date             | LSN     | Subject                                                                         | Readings (before class)                                                                                                                                                  | Due (at class start)                             |
|---------|------------------|---------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Block 1 | 6/7 Jan          | M1/T1   | Course Overview /<br>[1] Intro to Space Systems Engineering                     |                                                                                                                                                                          |                                                  |
|         | 9 Jan            | T2      | <b>T-DAY CLASS CANX (SNOW DAY)</b>                                              |                                                                                                                                                                          |                                                  |
|         | 8/13 Jan         | M2/T3   | [2] Requirements / Mission Design                                               | 7-15, 19-23, 53-55, 59-65                                                                                                                                                |                                                  |
|         | 12/15 Jan        | M3/T4   | [3] Mission Geometry / Constellation Design<br>/ Astrodynamics                  | 37-51, 56-58                                                                                                                                                             | Skills Review                                    |
|         | 14/20 Jan        | M4/T5   | [4] Embedded Systems<br><b>Lab 1: Introduction to Embedded Systems</b>          | Lab 1 Instructions                                                                                                                                                       | Lab 1 Tasks 0A/0B/0C                             |
|         | 16/22 Jan        | M5/T6   | [5] Metrology<br><b>Lab 2: Soldering</b>                                        | Lab 2 Instructions<br><a href="https://youtu.be/-qk-ulz05J8">https://youtu.be/-qk-ulz05J8</a><br><a href="https://youtu.be/eHu_fgAyNY4">https://youtu.be/eHu_fgAyNY4</a> |                                                  |
|         | 21/26 Jan        | M6/T7   | <b>Lab 1 Quiz</b><br>[6] Space Environment                                      | 71-82                                                                                                                                                                    | Lab 1 Email<br>(video/code/doc. statement)       |
|         | 23/28 Jan        | M7/T8   | [7] Electrical Power Subsystem (EPS)<br><b>Lab 3 Prelab Measurements</b>        | 151-162                                                                                                                                                                  | HW#1                                             |
|         | 27/30 Jan        | M8/T9   | [8] Communication Subsystem                                                     | 66-69<br>163-184                                                                                                                                                         |                                                  |
|         | 29 Jan           | M9      | <b>M-DAY CLASS CANX</b>                                                         |                                                                                                                                                                          |                                                  |
|         | 2/3 Feb          | M10/T10 | <b>Lab 3: Electrical Power Subsystem</b>                                        | Lab 3 Instructions                                                                                                                                                       | Lab 3 Pre-lab                                    |
|         | 4/5 Feb          | M11/T11 | <b>Lab 4: Communication Subsystem</b>                                           | Lab 4 Instructions                                                                                                                                                       |                                                  |
|         | 6/9 Feb          | M12/T12 | [9] Remote Sensing                                                              | 123-133                                                                                                                                                                  | HW #2                                            |
|         | 10/11 Feb        | M13/T13 | Analysis & Reporting / GR review                                                |                                                                                                                                                                          | Lab 4 Post-lab                                   |
| Block 2 | 12/13 Feb        | M14/T14 | <b>GR 1</b>                                                                     | LSN's [1] – [8]                                                                                                                                                          |                                                  |
|         | 17/18 Feb        | M15/T15 | <b>Lab 5: Remote Sensing</b>                                                    | Lab 5 Instructions                                                                                                                                                       | <b>Lab Report #1 (EPS)*</b>                      |
|         | 19/20 Feb        | M16/T16 | [10A] Attitude Determination and Control<br>Subsystem (ADCS) Part 1             | 185-198                                                                                                                                                                  | Lab 5 Post-Lab                                   |
|         | 23/24 Feb        | M17/T17 | <b>Lab 6: Attitude Determination</b>                                            | Lab 6 Instructions                                                                                                                                                       | Lab 6 Pre-lab                                    |
|         | 25/26 Feb        | M18/T18 | [10B] Attitude Determination and Control<br>Subsystem (ADCS) Part 2             |                                                                                                                                                                          | HW #3                                            |
|         | 27 Feb/<br>2 Mar | M19/T19 | <b>Lab 7: Attitude Control</b>                                                  | Lab 7 Instructions<br>Lab 8 Instructions                                                                                                                                 |                                                  |
|         | 3/4 Mar          | M20/T20 | [11] Thermal Control Subsystem (TCS)                                            | 213-221                                                                                                                                                                  | <b>Lab 8 Proposal*</b>                           |
|         | 5/6 Mar          | M21/T21 | <b>Lab 8: KestrelSAT Capstone Workday #1</b>                                    |                                                                                                                                                                          |                                                  |
|         | 9/10 Mar         | M22/T22 | [12] Budgets                                                                    | 97-104, 113-120                                                                                                                                                          | HW#4                                             |
|         | 11/12 Mar        | M23/T23 | [13] Fault Management and Reliability                                           | 83-95<br>233-239                                                                                                                                                         | <b>Lab Report #2 (ADCS)*</b>                     |
|         | 13/16 Mar        | M24/T24 | [14] Assembly, Integration, Test /<br>Verification, Validation                  | 223-231                                                                                                                                                                  |                                                  |
|         | 17/18 Mar        | M25/T25 | GR review                                                                       |                                                                                                                                                                          | HW #5                                            |
| Block 3 | 19/20 Mar        | M26/T26 | <b>GR 2</b>                                                                     | LSN's [9] – [14]                                                                                                                                                         |                                                  |
|         | 30/31 Mar        | M27/T27 | <b>Lab 8: KestrelSAT Capstone Workday #2</b>                                    |                                                                                                                                                                          |                                                  |
|         | 1/2 Apr          | M28/T28 | [15] Structures, Mechanisms, Manufacturing                                      | 135-149                                                                                                                                                                  |                                                  |
|         | 3/6 Apr          | M29/T29 | Vibration Demo & Astro Lab Tour<br><b>Lab 8: KestrelSAT Capstone Workday #2</b> |                                                                                                                                                                          |                                                  |
|         | 8 Apr            | T30     | [16] Propulsion Subsystem, Launch Systems,<br>Mission Operations                | 105-111, 199-212                                                                                                                                                         |                                                  |
|         | 9/10 Apr         | M31/T31 | <b>Field Trip</b>                                                               |                                                                                                                                                                          |                                                  |
|         | 13/14 Apr        | M32/T32 | <b>Lab 8 Presentations</b>                                                      |                                                                                                                                                                          | <b>Lab 8 Presentation<br/>(Advanced Topics)*</b> |
|         | 15/20 Apr        | M33/T33 | Review                                                                          |                                                                                                                                                                          | HW#6                                             |
|         | 21/22 Apr        | M34/T34 | <b>Final Exam - Part 1</b>                                                      |                                                                                                                                                                          |                                                  |
|         | 23/34 Apr        | M35/T35 | <b>Final Exam - Part 2</b>                                                      |                                                                                                                                                                          |                                                  |

\*Group turn-in